EE4-25
EE9-CS1-2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2015

MSc and EEE/EIE PART IV: MEng and ACGI

Correction
Q1. 10,20

# DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Monday, 18 May 10:00 am

Time allowed: 3:00 hours

Corrected Copy

**There are SIX questions on this paper.**

**Answer FOUR questions.**

*All questions carry equal marks*

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : E.C. Kerrigan

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1. Let the $n$-th order transfer matrix $G(s)$ have a state space realisation

$$G(s) \stackrel{s}{=} \left[\begin{array}{c|c} A & B \\ \hline C & 0 \end{array}\right]$$

and let

$$A'Q + QA + C'C = 0.$$

and

$$AP + PA' + BB' = 0$$

for some $Q = Q'$ and $P = P'$.

Suppose that

$$Q = \begin{bmatrix} Q_1 & 0 \\ 0 & 0 \end{bmatrix}.$$

and

$$P = \begin{bmatrix} P_1 & 0 \\ 0 & 0 \end{bmatrix}$$

where $\mathbb{R}^{n_1 \times n_1} \ni P_1 \succ 0$ and $\mathbb{R}^{n_1 \times n_1} \ni Q_1 \succ 0$ and where $n_1 < n$. Assume that $A$ has no eigenvalues on the imaginary axis.

a) Derive a decomposition of the system into its controllable/uncontrollable and observable/unobservable subsystems. The derivation should include proofs that the subsystems have the relevant properties. [ 8 ]

b) Prove that the controllable and observable subsystem is stable. [ 5 ]

c) State the conditions that guarantee that the system is stabilisable and detectable. [ 2 ]

d) Draw a diagram illustrating the subsystems of $G(s)$. [ 5 ]

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2. Consider the $\mathscr{H}_\infty$ filter for estimating $x$ shown in Figure 2.

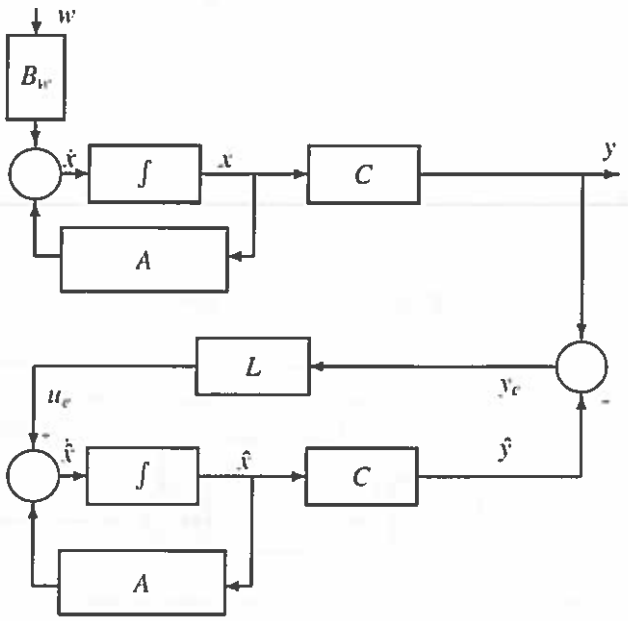



*Figure 2*

Let $z = x - \hat{x}$ and let $T_{zw}(s)$ denote the transfer matrix from $w$ to $z$. A stabilizing filter gain matrix $L$ is to be designed such that, for $\gamma > 0$, $\|T_{zw}\|_\infty < \gamma$.

a) Derive a state-space realisation for the transfer matrix $T_{zw}(s)$. [ 5 ]

b) Use the Bounded Real Lemma, stated in Question 5 below, to derive necessary and sufficient conditions for $\|T_{zw}\|_\infty < \gamma$. These conditions should be in the form of matrix inequality constraints. [ 10 ]

c) Suggest a transformation of variables that turns the matrix inequality derived in Part (b) above into a linear matrix inequality and write down this inequality. [ 5 ]

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3. Consider the feedback configuration in Figure 3. Here, $G(s)$ is a nominal plant model and $K(s)$ is a compensator. The stable transfer matrices $\Delta_a(s)$ and $\Delta_m(s)$ represent additive and multiplicative uncertainties on the nominal model.

The design specification are to synthesize a compensator $K(s)$ such that the feedback loop is internally stable when:

(i) $\Delta_a = 0$ and $\|\Delta_m(j\omega)\| \leq |w_m(j\omega)|, \forall\omega$, and,

(ii) $\Delta_m = 0$ and $\|\Delta_a(j\omega)\| \leq |w_a(j\omega)|, \forall\omega$.

where $w_a(s)$ and $w_m(s)$ are appropriate weighting functions.

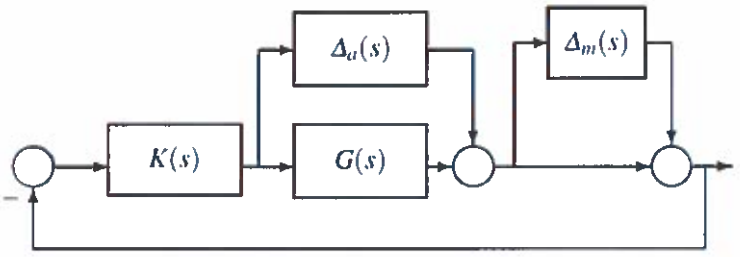



*Figure 3*

a) Derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w_a(s)$ and $w_m(s)$ that are sufficient to achieve the robust design specifications. [ 6 ]

b) Define suitable cost signals $z_1(s)$ and $z_2(s)$, external signal $w(s)$, measured signal $y(s)$ and control signal $u(s)$ and draw a block diagram, showing all these signals, as well as suitable weighting functions. [ 6 ]

c) Hence derive a generalised regulator formulation of the design problem that captures the sufficient conditions. [ 8 ]

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4. Consider the regulator in Figure 4 for which $x(0)=x_0$. A stabilizing state-feedback gain matrix $F$ is to be designed such that the cost function

$$J = \left\| \begin{bmatrix} u \\ z \end{bmatrix} \right\|_2^2$$

is minimized.

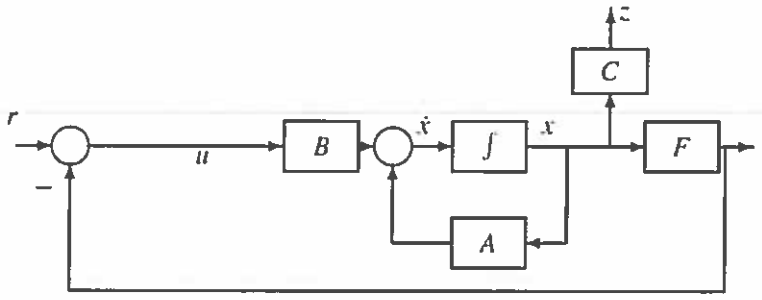



*Figure 4*

a) State the assumptions needed on the matrices $A$, $B$ and $C$ to obtain a solution and define a suitable Lyapunov function $V(t)$. [ 2 ]

b) Obtain an expression for $\int_0^\infty \dot{V}(t)dt$. State clearly any assumptions made. [ 2 ]

c) Find an expression for $F$ that minimizes $J$. Give also the algebraic Riccati equation that needs to be satisfied. What is the minimum value of $J$? [ 6 ]

d) Prove now that the closed loop system in Figure 4 is stable. State the assumptions required to guarantee stability. [ 4 ]

e) By evaluating an expression for $T_{zr}(s)$ in Figure 4 and using the Bounded Real Lemma (given in Question 5 below), prove that $\|T_{zr}\|_\infty < 1$. [ 6 ]

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5. a) Consider a state–variable model described by the dynamics

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

and let $H(s) = D + C(sI - A)^{-1}B$ denote the corresponding transfer matrix. Suppose there exists a $P = P' \succ 0$ such that

$$\begin{bmatrix} A'P + PA + C'C & PB + C'D \\ B'P + D'C & D'D - \gamma^2 I \end{bmatrix} \prec 0.$$

i) Prove that $A$ is stable. [ 4 ]

ii) By defining suitable Lyapunov and cost functions and completing a square, prove that

$$\|H\|_\infty < \gamma.$$

[ 6 ]

b) Consider the output injection problem shown in Figure 5. Note that the signal $w$ enters the system at two points. Let $T_{yw}(s)$ denote the transfer matrix from $w$ to $y$. An internally stabilizing output injection gain matrix $L$ is to be designed such that, for a given $\gamma > 0$, $\|T_{yw}\|_\infty < \gamma$.

i) Derive a state space realization for $T_{yw}(s)$. [ 4 ]

ii) By using the answer to Part (a) above, or otherwise, derive sufficient conditions for the existence of a feasible $L$. Your conditions should be in the form of the existence of certain solutions to linear matrix inequalities. [ 6 ]

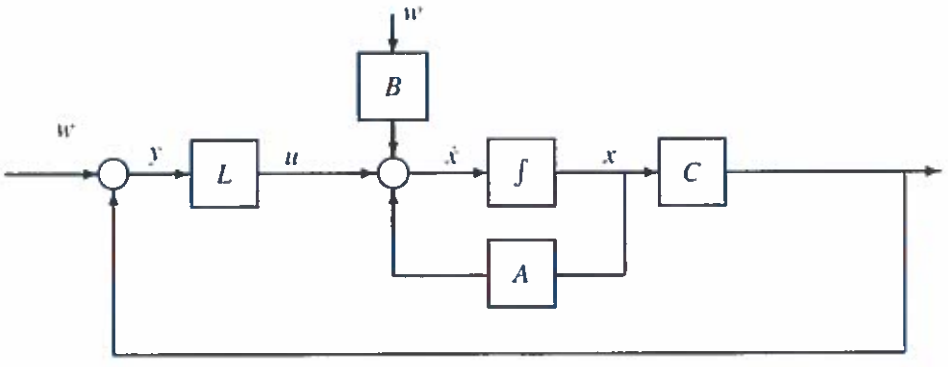



*Figure 5*

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6. Consider the regulator shown in Figure 6. Assume that

(i) The pair $(A,B)$ is controllable

(ii) $x(0)=0$

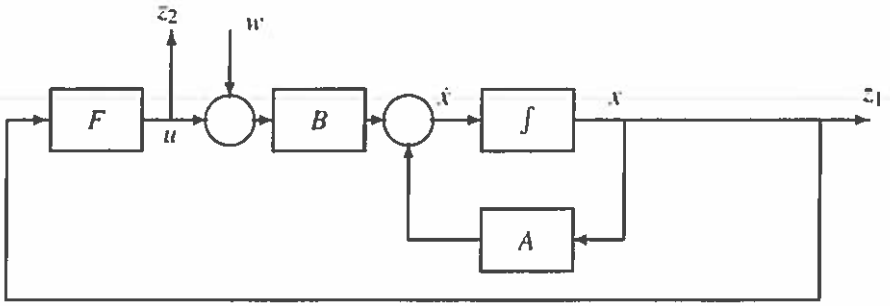



*Figure 6*

Let

$$z=\begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

and let $H(s)$ denote the transfer matrix from $w$ to $z$.

A stabilizing static output feedback gain matrix $F$ is to be designed such that, for $\gamma>0$, $\|H\|_\infty<\gamma$.

a) Write down the generalized regulator system for this design problem. [ 6 ]

b) By using the Lyapunov function $V(t)=x(t)'Xx(t)$, where $X$ is to be determined, derive sufficient conditions for the solution of the design problem.

Your conditions should be in the form of the existence of a certain solution to an algebraic Riccati equation.

It should also include an expression for $F$ and an expression for the worst-case disturbance $w$. [ 10 ]

c) Suppose that $A$ is stable. Show that $\gamma=1$ is feasible. [ 4 ]

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